

Nutrient leaching in undisturbed cores of an acidic sandy Podosol following simultaneous potassium chloride and di-ammonium phosphate application

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Abstract

In south-east Queensland, Australia, extensive areas of sandy soils (Podosols) with shallow (<1 m) watertables are used for exotic pine tree production. Despite concerns that surface-applied fertilisers (di-ammonium phosphate (DAP) and potassium chloride (KCl)) may be contributing to a decline in local groundwater quality, published information on nutrient leaching in these Podosols is scarce. Large (0.3 m i.d. × 0.85 m long) undisturbed soil cores were intermittently leached with deionised water following a single surface application of KCl in combination with DAP. Potassium was applied at rates (equivalent on a surface area basis) of 0 (K0), 50 (K50), 100 (K100) and 300 (K300) kg K⁺/ha, and DAP was applied at a rate equivalent to 50 kg P ha⁻¹. Applied ions appeared in the leachate very quickly after surface application, and reactive ions leached at the same rate as non-reactive ions. This behaviour was attributed to preferential soil–water flow, and limited ion sorption. About 30, 35, 100 and 25 percent of the applied K⁺, phosphorus (P), chloride (Cl⁻) and ammonium (NH₄⁺) was leached from the soil cores. Cation exchange was the major mechanism responsible for K⁺ and NH₄⁺ retention, although nitrification may have also contributed to NH₄⁺ losses. Findings indicate that significant amounts of surface-applied fertiliser ions can potentially be rapidly leached below the tree root-zone, and into the underlying groundwater. Immobile water regions were estimated to comprise nearly 50% of the soil–water. Intermittent leaching resulted in secondary concentration peaks along the trailing edge of the ion breakthrough curves (BTCs). This was attributed to diffusion of solute from immobile water regions into mobile water regions during periods of no-flow.

Introduction

Pollution of groundwater through leaching of surface-applied chemicals continues to be a worldwide topic of environmental concern (Nemati et al. 2003; Toride et al. 2003). This is particularly the case where fertilisers are applied to coarse-textured soils located in areas of moderate to high rainfall, and/or have shallow watertables

(Wang and Alva 2000). In south-east Queensland, Australia, exotic pine forest plantations have been established in many coastal areas. Soils of these areas are typically coarse-textured, acidic (pH 3–4), possess limited nutrient retention properties, and have shallow (<1 m) watertables that are hydraulically connected to environmentally-sensitive marine environments. Consequently, the leaching potential of these soils is high.

During plantation establishment, seedlings receive fertiliser additions of 50–100 kg K⁺ ha⁻¹ as potassium chloride (KCl) and 50 kg P ha⁻¹ as di-ammonium phosphate (DAP), and there is concern that leaching of fertiliser ions could be contributing to local groundwater and marine pollution. The simultaneous addition of K⁺ and NH₄⁺-based fertilisers can further increase the leaching potential through competition for the limited exchange sites (Lumbanraja and Evangelou 1992; Chappell and Evangelou 2000). Furthermore, the addition of K⁺ and NH₄⁺-based fertilisers can displace resident exchangeable cations (e.g. calcium Ca²⁺) and magnesium (Mg²⁺), thereby increasing their susceptibility to leaching (Phillips et al. 1988a, b; Johns and Vimpany 1999). Movement of inorganic phosphorus (P) is extremely limited in many fine-textured soils, but the mobility of this anion can increase dramatically in acidic, coarse-textured sandy soils (Gerritse 1989; Chen et al. 1996) such as Podosols, and in well-structured soils exhibiting preferential flow through continuous macropores (e.g. Jensen et al. 2000).

Little published information on ion (K⁺, NH₄⁺, P, nitrate, (NO₃⁻), Ca²⁺, Mg²⁺, Na⁺ and Cl⁻) leaching in these coastal sands is available, and as such, their environmental impact cannot be quantified. Therefore, more information on fertiliser leaching in these coarse-textured soils is required to ensure appropriate management strategies are developed for protecting the ecologically-sensitive receiving soil and groundwater bodies. The objective of this study was to investigate nutrient leaching in large, undisturbed cores of a coarse-textured sand currently used for pine tree production.

Materials and methods

Leaching studies

Sampling location and initial soil properties

Samples of a coarse-textured Podosol soil (Isbell 1996), or a Podzol (Soil Survey Staff 1975), were collected from a second rotation exotic pine forest plantation on Bribie Island, Queensland, Australia (27° 00.685' S 153° 09.722' E). The site was relatively flat, and was located in a region receiving an annual rainfall of approximately 1323 mm.

A perched groundwater system was located approximately 0.85 m below the ground surface. This groundwater was hydraulically connected to the overlying unsaturated soil, and groundwater discharges to nearby Moreton Bay and the Pumicestone Passage.

Grab samples were collected from the 0–30, 30–50 and > 50 cm depths to determine initial soil chemical and physical properties (Table 1). Prior to analysis, samples were air-dried and sieved (<2 mm) to remove larger pieces of woody material. Unless otherwise stated, the analytical procedures used were those described in Rayment and Higginson (1992). Exchangeable Al³⁺, Ca²⁺, Mg²⁺, K⁺, and Na⁺ were extracted by shaking samples of soil with 0.01 M silver thiourea (AgTU) at a soil to solution ratio of 1:20 for 2 h (Method 15F1), and the extract analysed by Atomic Absorption Spectroscopy (AAS; Varian Spectra AA). Available-P was determined using 0.5 M sodium bicarbonate (NaHCO₃ pH 8.5) at a soil to solution ratio of 1:100 (Colwell P; Method 9B1). Extracts were analysed manually for P using the molybdenum blue method. Total P (TP) was determined by sulfuric acid/potassium sulfate digestion and FIA techniques (QuickChem method 13-115-01-1-B). Total N (TN) and Total Carbon (TC) were determined using mass spectrometry techniques (Isoprime Isotope Ratio Mass Spectrometer). Inorganic N (NH₄⁺-N and NO₃⁻-N) was extracted using 2 M KCl (1:20 soil to solution ratio for 1 h; Method 7C1) and the extracts were analysed using FIA techniques (QuickChem methods 10-107-06-4-D and 10-107-04-1-H). Soil pH (Milli-Q water and 0.01 M CaCl₂; Methods 4A1 and 4B2 respectively) and electrical conductivity (EC) (Method 3A1) were determined using a soil to water ratio of 1:5. Hydrous Fe and Al hydroxides were extracted using ammonium oxalate (McKeague and Day 1966). Soil–water retention curves were determined as described by Klute (1986). Soil texture was estimated using the hydrometer method (Gee and Bauder 1986). Soil mineralogy was determined by powder X-ray diffraction analysis using a Philips Vertical Diffractometer employing Co-K α radiation. Permanent surface charge (σ_p) was measured using the cesium adsorption method (Anderson and Sposito 1991). Field saturated hydraulic conductivity (k_{sat}) was measured using a Guelph constant head permeameter.

Table 1. Selected properties of the Podosol soil.

Parameter	0–30 cm	30–50 cm	> 50 cm
ρ_b (g cm ⁻³)	1.0	1.6	1.6
pH	3.34	3.45	3.59
EC (μ S cm ⁻¹)	118	187	14
k_{sat} ($\times 10^{-5}$ m s ⁻¹)	13.7	24.5	26.3
Total N (mg kg ⁻¹)	405	102	30
2 M KCl NH ₄ ⁺ -N (mg kg ⁻¹)	7.18	0.39	0.22
2 M KCl NO ₃ ⁻ -N (mg kg ⁻¹)	0.86	0.87	0.62
Total P (mg kg ⁻¹)	28.87	12.95	3.55
0.5 M NaHCO ₃ P (mg kg ⁻¹)	13.64	3.98	0.94
Total C (%)	1.48	0.27	0.05
Mineralogy	Quartz ($\approx 100\%$)	Quartz ($\approx 100\%$)	Quartz ($\approx 100\%$)
Hydrous Fe oxide (mg kg ⁻¹)	190	96	50
Hydrous Al oxide (mg kg ⁻¹)	220	121	40
Texture	Sand	Sand	Sand
Exchangeable cations (cmol _c kg ⁻¹)			
Ca ²⁺	1.123	0.194	0.005
Mg ²⁺	0.239	0.152	0.023
K ⁺	0.012	0.005	0.001
Na ⁺	0.008	0.004	0.001
Al ³⁺	0.735	0.556	0.140
ECEC* (cmol _c kg ⁻¹)	2.117	0.911	0.170
σ_p (cmol _c kg ⁻¹)	0.73	0.19	0.16

*ECEC = Effective Cation Exchange Capacity = Summation of exchangeable cations.

Collection and automation of undisturbed soil core

Four (4) undisturbed cores (30 cm i.d. $\times$ 100 cm long) were collected as outlined by Phillips (2001). On return to the laboratory, a tension drainage system was attached to the base of each soil core. This system maintained a constant vacuum (10 kPa) to avoid saturated/ponded conditions at the base of the soil core during each leaching event. Soil water tension and volumetric water content throughout the leaching experiment were monitored at depths of 15, 40 and 65 cm in each soil core. Soil water tension was monitored using Jet-fill tensiometers (Soil Moisture Equipment Corp., Santa Barbara, CA: #652X03-B1M1), which had been calibrated over the range of 0–80 kPa. Volumetric water content was monitored using Thetaprobe Soil Moisture Sensors (Delta-T Devices Ltd., Cambridge, England; Model #ML1). Each thetaprobe had been calibrated over a water content range of 0–0.35 m³ m⁻³ using material from the 0–30, 30–50 and > 50 cm depths. Tensiometers and thetaprobes were inserted horizontally through the side of each soil core, and the voltage outputs recorded with a DT500 Datalogger at 3 min intervals (Phillips 2001).

Fertiliser treatments and leaching procedure

Prior to fertiliser application, each soil core was leached with approximately 1 pore volume (21 l) of deionised water over a 4-month period to remove much of the water-soluble ions. The imposed fertiliser treatments (equivalent on a surface area basis) were applied to the soil surface in 100 ml of solution containing either: 0 kg K⁺ ha⁻¹ (as KCl) + 50 kg P/ha (as DAP); 50 kg K⁺ ha⁻¹ + 50 kg P ha⁻¹; 100 kg K⁺ ha⁻¹ + 50 kg P ha⁻¹; and 300 kg K⁺ ha⁻¹ + 50 kg P ha⁻¹ (treatment K0, K50, K100 and K300). After application, the fertiliser solution was allowed to react with the soil for 2-h prior to commencing leaching with deionised water.

Each soil core was intermittently leached with deionised water using a calibrated rainfall simulator. A Masterflex peristaltic pump (fitted with a dual pump head) delivered water to each rainfall simulator, which supplied water to the soil surface at a constant flux density (2.1 cm h⁻¹). Over the duration of the experiment, each soil core underwent 10 leaching events, and received approximately 65 cm of irrigation (equivalent to 50% of average annual rainfall). The experiment was terminated after the fertiliser pulse had emerged from

the base of the soil core, and the subsequent EC of the leachate remained relatively low and constant ($\approx 50 \mu\text{S cm}^{-1}$).

Evaporation (E) losses were monitored throughout the experiment on a separate core (30 cm i.d. $\times$ 10 cm long) packed with material from the 0–30 cm depth interval. During the early times ($t < 250$ h), the evaporation rate was relatively constant ($0.00383 \text{ cm h}^{-1}$; $r^2 = 0.996$). For longer time periods (1000 h), the rate of loss followed a logarithmic form ($E = 0.497 \times \ln(t) - 1.723$; $r^2 = 0.95$).

Leachate collection and analysis

Leachate was collected in aliquots of about 500 ml. The volume of each aliquot was estimated by weighing. About 50 ml of leachate were removed and analysed immediately for pH and EC. Approximately 200 ml of leachate were passed through a $0.45 \mu\text{m}$ filter. Samples of filtered leachate were placed into polycarbonate containers and frozen until required for analysis. Unless otherwise stated, the analytical procedures used were those described in Rayment and Higginson (1992). Leachate samples were analysed for pH and EC using appropriate meters; Al^{3+} , Ca^{2+} , Mg^{2+} , K^+ , and Na^+ by AAS; Cl^- (QuickChem method 10-117-07-2-B), molybdate reactive phosphate (MRP) (QuickChem method 31-115-01-3-A), NH_4^+ -N (QuickChem method 10-107-06-4-D), and NO_3^- -N (QuickChem method 10-107-04-1-H) by flow injection analysis (LaChat Flow Injection Analysis QuickChem 8000). Hereafter, MRP, NH_4^+ -N, and NO_3^- -N will be referred to as P, NH_4^+ and NO_3^- , respectively.

Potassium, NH_4^+ and P sorption isotherms

Potassium, NH_4^+ and P sorption isotherms ($n = 2$) were determined for materials from the 0–30, 30–50 and > 50 cm depths using the standard batch technique (Roy et al. 1991). Nine (9) initial solutions were made by diluting the input solution for K300 (see above), and these solutions contained 0–300 mg $\text{K}^+ \text{ l}^{-1}$, 0–50 mg P l^{-1} and 0–45 mg $\text{NH}_4^+ \text{ l}^{-1}$. Soil (approximately equivalent to 5 g of dry material) was weighed into a series of 50 ml polypropylene centrifuge tubes, and 25 ml of solution added. The suspension was shaken end-over-end for 16 h, then centrifuged at 3000 rpm for

30 min, and passed through a $0.45 \mu\text{m}$ filter prior to analysis. The initial and equilibrium solutions were analysed for K^+ , P, NH_4^+ , Ca^{2+} , Al^{3+} , Mg^{2+} and Na^+ as stated above, and the difference between these two values used to calculate the amount of each ion sorbed by the soil.

Potassium and NH_4^+ exchange behaviour were interpreted using the Quantity (Q)/Intensity (I) approach (Beckett 1964a, b), in which the change in exchangeable K^+ or NH_4^+ ($\text{cmol}_c \text{ kg}^{-1}$) is plotted against its corresponding solution activity ratio (AR_K or AR_NH_4). The activity ratio is commonly expressed as:

$$\text{AR}_\text{K} = (\text{K}^+) \div (\text{Ca}^{2+} + \text{Mg}^{2+})^{1/2} \quad (1a)$$

$$\text{AR}_\text{NH}_4 = (\text{NH}_4^+) \div (\text{Ca}^{2+} + \text{Mg}^{2+})^{1/2} \quad (1b)$$

where () denotes cation activities in the equilibrium solution. In acid soils, however, Equation 1 has been modified to include the contribution of Al^{3+} (Tinker 1964; Phillips et al. 1988a):

$$\text{AR}_\text{K} = (\text{K}^+) \div \{(\text{Al}^{3+})^{1/3} + (\text{Ca}^{2+} + \text{Mg}^{2+})^{1/2}\} \quad (2a)$$

$$\text{AR}_\text{NH}_4 = (\text{NH}_4^+) \div \{(\text{Al}^{3+})^{1/3} + (\text{Ca}^{2+} + \text{Mg}^{2+})^{1/2}\} \quad (2b)$$

Other important parameters obtained from a Q/I plot are: the potential buffering capacity (PBC_K or PBC_NH_4 , $\text{cmol}_c \text{ kg}^{-1} \text{ mol l}^{-1}$), represented by the slope of the linear part of the Q/I plot; the labile portion ($-\Delta\text{K}^0$ or $-\Delta\text{NH}_4^0 \text{ cmol}_c \text{ kg}^{-1}$) obtained by extrapolating the linear part of the Q/I plot to AR_K or $\text{AR}_\text{NH}_4 = 0$; the concentration of K^+ or NH_4^+ retained by specific adsorption sites (K_sas or NH_4sas , $\text{cmol}_c \text{ kg}^{-1}$), and the equilibrium activity ratio ($\text{AR}_\text{o}^\text{K}$ or $\text{AR}_\text{o}^\text{NH}_4$) represented by the AR at which the change in exchangeable K^+ ($\Delta\text{ExK}^+ \text{ cmol}_c \text{ kg}^{-1}$) or NH_4^+ ($\Delta\text{ExNH}_4^+ \text{ cmol}_c \text{ kg}^{-1}$) was zero.

Potassium (k_K) and NH_4^+ (k_NH_4) selectivity coefficients were calculated using the Gapon relationship (Sposito 2000), modified to include the effects of Al^{3+} , as (Phillips et al. 1988a):

$$[\text{K}^+] \div [\text{Al}^{3+} + \text{Ca}^{2+} + \text{Mg}^{2+}] = k_\text{K} \times \text{AR}_\text{K} \quad (3a)$$

$$[\text{NH}_4^+] \div [\text{Al}^{3+} + \text{Ca}^{2+} + \text{Mg}^{2+}] = k_{\text{NH}_4} \times \text{AR}_{\text{NH}_4} \quad (3b)$$

where $[\]$ denotes the cation concentration in the exchange phase ($\text{cmol}_c \text{ kg}^{-1}$).

Equation 1 cannot describe P behaviour in soil because the release of sorbed P into solution is not completely reversible, and the Freundlich equation has been commonly used to describe chemisorption isotherms (McBride 2000). The Freundlich equation has the form (McBride 2000):

$$S = kc \quad (4)$$

where S is the increase in sorbed P per unit weight of soil (mg kg^{-1}), c is the final P solution concentration (mg l^{-1}), k_f and n are empirical fitting constants. The parameter k_f can be used as a sorption index, while the parameter n can be likened to a bonding strength as it determines the overall shape of the sorption curve (Sanyal and De Datta 1991).

Results and discussion

Potassium, NH_4^+ and P sorption isotherms

Potassium and NH_4^+ Q/I plots for the 0–30 and 30–50 cm depths are presented in Figure 1a and b respectively. The estimated concentration of K^+ specific sorption sites (K_{sas}) for the 0–30, 30–50 and > 50 cm material were 0.0034, 0.0014 and 0.0019 $\text{cmol}_c \text{ kg}^{-1}$, respectively, and the potential K^+ buffering capacity (PBC_K) was 7.17, 5.04 and 2.95 $\text{cmol}_c \text{ kg}^{-1} \text{ mol l}^{-1}$, respectively (Table 2). The concentrations of K^+ specific sorption sites and potential buffering capacity for this sandy soil were considerably lower than those values reported for silty (0.13–0.91 $\text{cmol}_c \text{ kg}^{-1}$ and 26.2–79.8 $\text{cmol}_c \text{ kg}^{-1} \text{ mol l}^{-1}$) and clayey (0.88–1.22 $\text{cmol}_c \text{ kg}^{-1}$ and 93.6–151.5 $\text{cmol}_c \text{ kg}^{-1} \text{ mol l}^{-1}$) soils (Evangelou et al. 1986; Evangelou and Blevins 1988; Lumbanraja and Evangelou 1992). The estimated concentration of NH_4^+ specific sorption sites ($\text{NH}_{4\text{sas}}$) for the 0–30, 30–50 and > 50 cm material were 0.0075, 0.0048 and 0.0019 $\text{cmol}_c \text{ kg}^{-1}$, respectively, and the potential NH_4^+ buffering capacity (PBC_{NH_4}) was 3.86, 3.23 and 1.95 $\text{cmol}_c \text{ kg}^{-1} \text{ mol l}^{-1}$, respectively (Table 2). These values are similar to those calculated for K^+ , and those reported by Wang and Alva

(2000) for two sandy soils ($\text{NH}_{4\text{sas}} = 0.0027$ –0.0096 $\text{cmol}_c \text{ kg}^{-1}$; $\text{PBC}_{\text{NH}_4} = 0.13$ –3.9 $\text{cmol}_c \text{ kg}^{-1} \text{ mol l}^{-1}$), but were much lower than those reported by Thompson and Blackmer (1992) for clay loam soils ($\text{NH}_{4\text{sas}} = 0.0027$ –0.0096 $\text{cmol}_c \text{ kg}^{-1}$). The low buffering capacity of this Podisol compared with reported values for clay soils is probably due to the Podisol's very low cation exchange capacity ($\text{ECEC} < 2.2 \text{ cmol}_c \text{ kg}^{-1}$; Table 1). Various workers have reported a positive linear relationship between PBC and cation exchange capacity for K^+ and NH_4^+ (Goedert et al. 1975; Lumbanraja and Evangelou 1990; Pal et al. 1999), and a similar finding was observed for the soil materials used in this study (PBC_{NH_4} : $r^2 = 0.94$ and PBC_K : $r^2 = 0.99$).

The near-linear shape of the Q/I plots for both cations in the 0–30 and 30–50 cm depths, coupled with the very small concentration of specific sorption sites, suggests that both K^+ and NH_4^+ sorption occurred predominantly via cation exchange reactions. The importance of this reaction was supported by increasing concentrations of resident Al^{3+} , Ca^{2+} , Mg^{2+} and Na^+ in the equilibrium solution with increasing concentrations of added K^+ and NH_4^+ . The very small number of specific sorption sites in this Podisol soil can be attributed to the absence of micaceous clay minerals with an interlayered structure (e.g. illite and vermiculite), and kaolinite, which have a high affinity for K^+ ions (Pleysier et al. 1979; Levy et al. 1988; Phillips et al. 1988a; Barbayannis et al. 1996; Chappell and Evangelou 2000).

In contrast to the 0–30 and 30–50 cm depths, K^+ and NH_4^+ Q/I plots for the > 50 cm depth tended to be non-linear in shape (Figure 1a and b). Non-linearity has often been attributed to the presence of highly specific sorption sites (e.g. Phillips et al. 1988a); however, this mechanism would not explain the non-linear sorption observed for this Podisol given the negligible concentration of specific sorption sites for both K^+ and NH_4^+ (Table 2). It is more likely that both K^+ and NH_4^+ sorption was restricted by a lack of available cation exchange sites at the higher solution activity ratios (i.e. $\text{AR}_K > 0.04$ and $\text{AR}_{\text{NH}_4} > 0.015$). For example, for $\text{AR}_K < 0.04$, the change in exchangeable K^+ increased nearly linearly with increasing AR_K to a concentration of about 0.17 $\text{cmol}_c \text{ kg}^{-1}$ (Figure 1a), and this represented about 100% of the ECEC. However, for

Table 2. Parameters obtained from Q/I plots for K^+ and NH_4^+ for each soil depth.

Parameter	Ion	Soil depth (cm)		
		0–30	30–50	> 50
Specific sorption sites ($\times 10^{-3}$ cmol _c kg ⁻¹)	K^+	3.4	1.4	1.9
PBC (cmol _c kg ⁻¹ mol l ⁻¹)	K^+	7.17	5.04	2.95
AR ₀ ($\times 10^{-4}$ mol l ⁻¹)	K^+	3.5	3.2	1.4
Gapon selectivity coefficient (l mol ⁻¹) (mean $\pm$ standard deviation)	K^+	5.5 ± 1.2	5.2 ± 0.9	14.2 ± 2.5
Specific sorption sites ($\times 10^{-3}$ cmol _c kg ⁻¹)	NH_4^+	7.5	4.8	1.9
PBC (cmol _c kg ⁻¹ mol l ⁻¹)	NH_4^+	3.86	3.23	1.96
AR ₀ ($\times 10^{-4}$ mol l ⁻¹)	NH_4^+	3.4	2.4	1.5
Gapon selectivity coefficient (l mol ⁻¹) (mean $\pm$ standard deviation)	NH_4^+	1.5 ± 0.8	4.6 ± 1.6	10.8 ± 4.1

$AR_K > 0.04$, no further increase in exchangeable K^+ was observed since the sorption sites were saturated with K^+ and no additional sorption sites were available. This conclusion is further supported by (a) the negligible change in the K^+ and NH_4^+ exchange ratio for $AR_K > 0.04$ and $AR_{NH_4} > 0.015$, respectively (Figure 1c and d), and (b) the absence of this saturation effect in the 0–30 and 30–50 cm depths due to their greater ECEC (Table 1).

Potassium sorption was generally found to be greater than NH_4^+ sorption, and sorption for both cations was greatest by the 0–30 cm material and least by the > 50 cm material. Often the magnitude of cation sorption via exchange reactions increases with increasing concentration of negative surface charge (Phillips et al. 1988a), and this trend was observed in this study (Figure 1a and b and Table 1). However, sites with a stronger selectivity for one cation relative to another will also affect the magnitude of sorption. The effect of AR on the Gapon selectivity coefficient for K^+ (k_K) and NH_4^+ (k_{NH_4}) is presented in Figure 1e and f. Potassium selectivity coefficients demonstrated little variation with AR_K , with a mean value of 5.5 ± 1.2 , 5.2 ± 0.9 and 14.2 ± 2.5 l mol⁻¹ for the 0–30, 30–50 and > 50 cm depths, respectively (Table 2), and these values are within the range reported by Pal et al. (1999) for a range of siliceous sands (0.04–10.5 l mol⁻¹) from Australia. Values of k_{NH_4} tended to decrease with increasing AR_{NH_4} and then remained relatively constant at higher AR, as observed in previous studies (e.g. Levy et al. 1988; Phillips 1988a). Average values of k_{NH_4} were 1.5 ± 0.8 , 4.6 ± 1.6 and 10.8 ± 4.1 l mol⁻¹ for the 0–30, 30–50 and > 50 cm depths, respectively (Table 2). Clearly, the > 50 cm depth displayed much greater

selectivity for K^+ and NH_4^+ relative to the 0–30 and 30–50 cm depths, but the extent of sorption was severely restricted by the very low ECEC (Table 1), particularly at $AR_K > 0.04$ and $AR_{NH_4} > 0.015$ (Figure 1e and f).

The permanent charge (σ_p) component represents 34%, 21% and 94% of the ECEC for the 0–30, 30–50 and > 50 cm depth materials respectively (Table 1). The remainder of the ECEC was assumed to arise from variable charge sites on organic matter and to a lesser extent, hydrous Fe and Al oxides (Table 1). Both of these sources of charge exhibit a stronger preference for multi-valent cations relative to monovalent cations (e.g. Pleyrier et al. 1979; Phillips et al. 1988a; Barbayiannis et al. 1996). Therefore, the higher concentrations of organic matter and hydrous Fe and Al oxides in the 0–30 and 30–50 cm depth materials may explain the lower k_K and k_{NH_4} for these materials relative to that for the > 50 cm depth.

Interestingly, k_K tended to be greater than k_{NH_4} for all soil materials (Table 2). This indicates that these two cations did not behave as true exchange analogs as commonly believed given their similar ionic radii and valence, and for this study, the Podsol appears to have a stronger selectivity for K^+ relative to NH_4^+ . Although Barbayiannis et al. (1996) came to the same conclusion studying competitive K^+ and NH_4^+ sorption by soils rich with micaceous clay minerals, these workers found that NH_4^+ was more capable of displacing K^+ from the exchange sites. The contrasting sorption and selectivity behaviour of K^+ and NH_4^+ observed by Barbayiannis et al. (1996) to that found in this study demonstrates the complex nature of cation reactions in soil, and highlights the need for continued research into one of the most fundamental chemical process occurring in all soils.

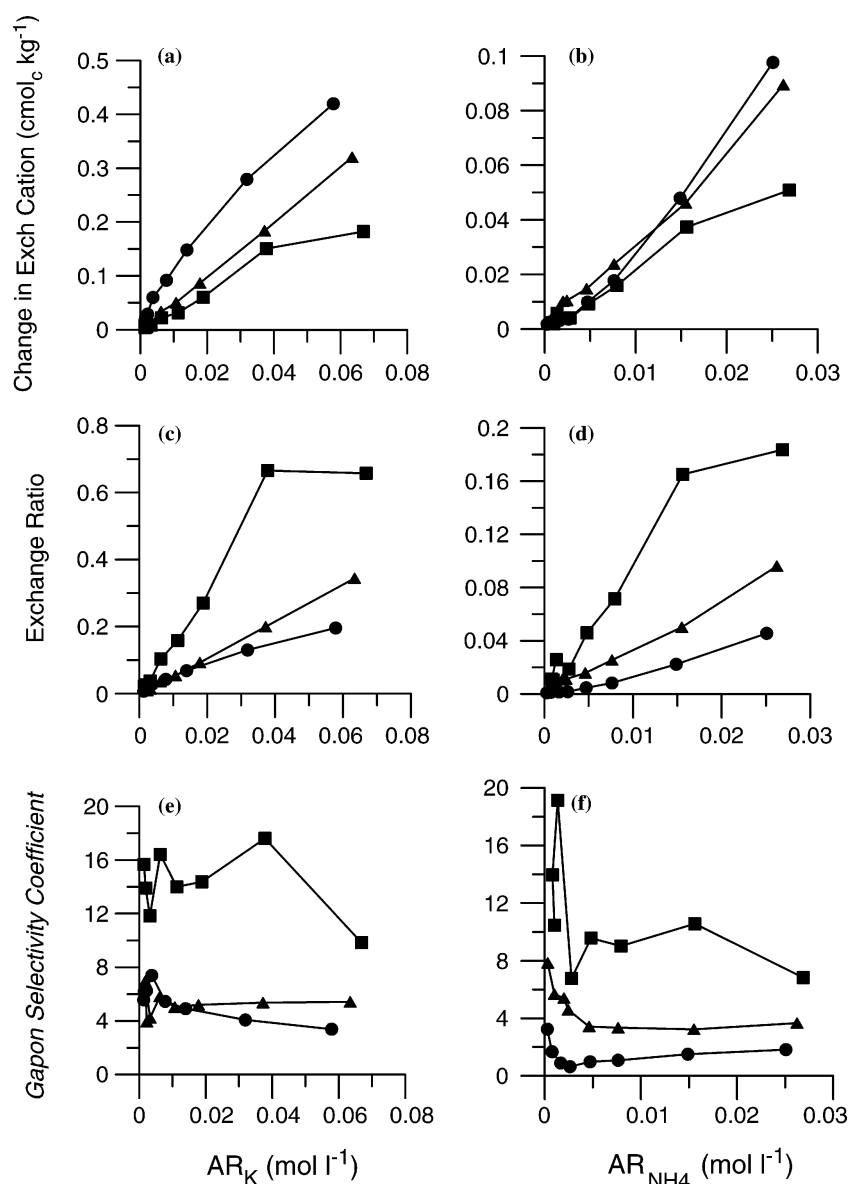


Figure 1. (a) Potassium and (b) NH₄⁺ Q/I plots, (c) K⁺ and (d) NH₄⁺ exchange ratio plots, and (e) K⁺ and (f) NH₄⁺ Gapon selectivity coefficients for the 0–30 cm (●), 30–50 cm (▲) and > 50 cm (■) materials.

All soil materials displayed a poor ability to sorb added P (Figure 2), and this was attributed to the lack of P sorbing materials such as hydrous iron and aluminium oxides and clay minerals with positively charged sites (e.g. edge sites of kaolinite; Table 1), and the competitive effects of dissolved organic ligands (Sanyal and De Datta 1991; Jensen et al. 2000; Phillips 2002a, b). Dissolved organic acids may also form complexes with metal cations (e.g. Fe and Al) in which dissolved inorganic P may participate as a ligand (Gerke and Hermann

1992; Jensen et al. 2000), thereby maintaining higher solution inorganic P concentrations. On the other hand, organic matter sorbed to soil particles can also remove a proportion of soluble inorganic P by its interaction with organo-metal complexes (Sanyal and De Datta 1991). The higher organic matter and hydrous Fe and Al oxide contents of the 0–30 cm depth material (Table 1) may explain why it sorbed relatively more added P than the other samples. Although the mechanism of P sorption was not quantified in this study, the

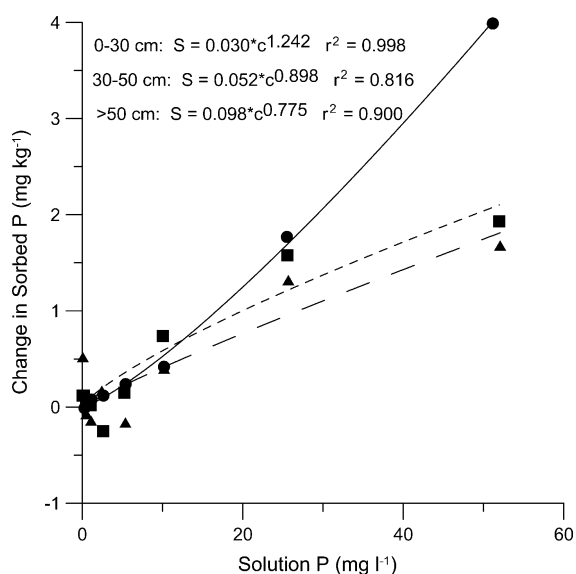


Figure 2. Phosphorus sorption isotherm for the 0–30 cm ($\blacktriangle$), 30–50 cm ($\blacksquare$) and > 50 cm ($\bullet$) materials and corresponding Freundlich parameters (Equation 4).

isotherms clearly indicate that P leaching may be of concern in this Podsol soil.

Phosphorus sorption isotherms were adequately described by the Freundlich equation, and the resulting sorption parameters are given in Figure 2. Values for k_f ranged from 0.030–0.098 l kg^{-1} , and values for n ranged from 0.775 to 1.242. These values are typically much lower than those reported for other sandy and sesquioxide-rich clay soils (Chen et al. 1996; Phillips 2002a). For example, k_f values for P sorption by sandy soils have been reported as between 6.7–67.6 l kg^{-1} , and between 38–281 l kg^{-1} for sesquioxide-rich clay soils. Clearly, the acidic nature and absence of appreciable amounts of hydrous iron and aluminium oxides highlights the high P leaching potential in the Podsol soil.

Leaching experiments

Water movement

Changes in soil–water content (θ_v) as a function of time and depth in Cores 1 and 4 are presented in Figure 3a. This pattern was observed in all cores and for each leaching event, although the time required for the wetting front to reach the position of each thetaprobe decreased with increasing

initial soil–water content (θ_i), particularly in the surface soil zone (0–15 cm). The time required for the wetting front to reach each thetaprobe provided an excellent approximation of the average *in situ* pore-water velocity (v). Values of v were between 40 and 110 cm h^{-1} in the 0–15 cm interval, and about 40 cm h^{-1} in the 15–85 cm depth interval (Figure 3b). These high values of v , coupled with the soil's poor water retention characteristics (Figure 3d), resulted in up to 90% of the surface-applied water draining from the soil cores per leaching event (Figure 3c).

The rapid movement of applied water in this coarse-textured (sandy) soil suggests that preferential flow to some extent was occurring. This was not due to wall or edge flow as the thetaprobe sensors extended 10 cm into the soil core, and they produced a steady increase in readings as would be expected with water flow through soil (Figure 3a). Other sources of preferential flow include the presence of immobile/mobile soil–water regions (Nkedi-Kizza et al. 1982), and water repellency (Ritsema and Dekker 1996). The degree of water repellency of the 0–30, 30–50 and > 50 cm depth materials, as well as the surface of each soil core, was determined using the water drop penetration time (WDPT) procedure (Ritsema and Dekker 1996). The 30–50 and > 50 cm depths and the soil core surface, were classified as wettable (non-water repellent), whereas the 0–30 cm depth was classified as slightly-water repellent. The difference between the 0–30 cm material and the soil core surface (as these are the same materials) may be attributed to the lower water content of the 0–30 cm sample ($\theta_v = 0.01 \text{ cm}^3 \text{ cm}^{-3}$) relative to that of the soil core surface ($\theta_v = 0.08 \text{ cm}^3 \text{ cm}^{-3}$). Ritsema and Dekker (1996) cite numerous workers that found water repellency disappears with increasing soil–water content. Furthermore, water repellency is most obvious in highly organic soils (> 5% organic matter) (Ritsema and Dekker 1996; McKissock et al. 1998). Since the Podsol did not exhibit water repellent characteristics (low WDPT and low organic matter (Table 1)) the rapid movement of applied water is not attributed to water repellency. The possibility of immobile soil–water regions is discussed below.

Leachate breakthrough curves (BTCs) for Ca^{2+} , Mg^{2+} , K^+ , Na^+ , NH_4^+ , NO_3^- , Cl^- and P are presented in Figure 4. Leaching below a depth of 85 cm was evident for all applied ions

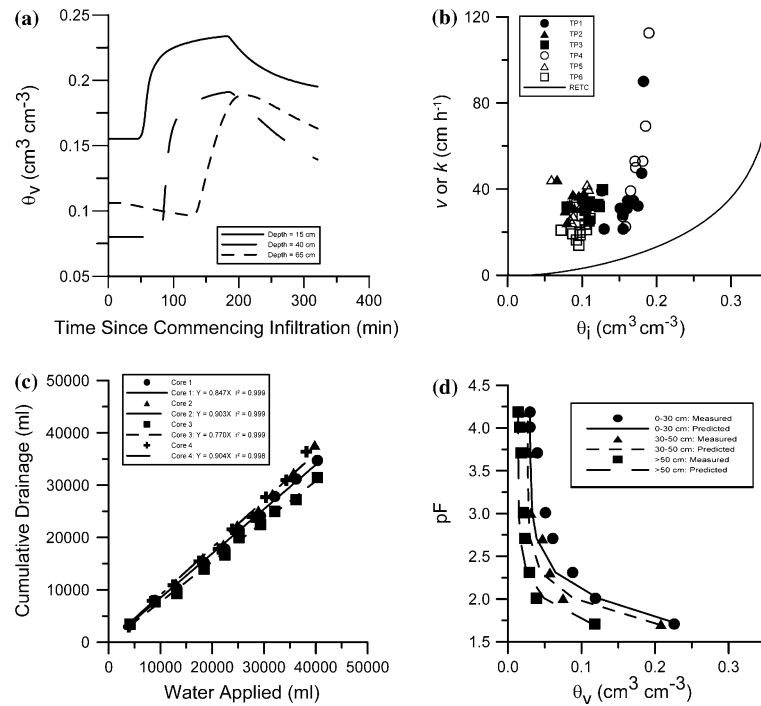


Figure 3. (a) Measured changes in volumetric soil-water content (θ_v) as a function of time at depths of 15, 40 and 65 cm; (b) effect of initial soil-water content (θ_i) on *in situ* pore-water velocity as measured by thetaprobes 1–6 (TP1-TP6) and predicted hydraulic conductivity using RETC; (c) measured and predicted relationship between applied water and cumulative drainage for each soil core; and (d) measured and predicted (RETC) soil moisture retention curves for the 0–30, 30–50 and >50 cm depths.

(K^+ , Cl^- , NH_4^+ and P) and ions initially present within the soil core (i.e. Ca^{2+} , Mg^{2+} and Na^+), and the magnitude of these losses increased with increasing rate of KCl application (Table 3). The presence of Ca^{2+} , Mg^{2+} and Na^+ in the leachate is considered to be a result of cation exchange between the applied K^+ (and NH_4^+ to a lesser extent) and resident exchangeable Ca^{2+} , Mg^{2+} and Na^+ (see previous section Potassium, NH_4^+ and P Sorption Isotherms). This mechanism is commonly observed in leaching studies involving macro-nutrient cations such as Ca^{2+} , Mg^{2+} , K^+ and Na^+ (Pleysier and Juo 1981; Swarup et al. 1983; Phillips et al. 1988b; Phillips 2002b, 2004).

With the exception of NO_3^- , the distribution of the ions in the leachate displayed the following behaviour. Prior to fertiliser application, ionic concentrations decreased with increasing volume of drainage due to progressive removal of resident soluble ions. This trend continued for a short period after fertiliser addition, after which leachate concentrations rapidly increased to a maximum (leading edge) and thereafter decreased (trailing

edge). All ions reached a maximum concentration after approximately 5000 ml (equivalent to 0.25 pore volumes) of drainage, showing that the cations (Ca^{2+} , Mg^{2+} , K^+ , Na^+ and NH_4^+) and anions (Cl^- and P) leached at essentially the same rate. This implies that cation exchange and/or other sorption processes did not have a major affect on the rate of movement of K^+ , NH_4^+ and P relative to the average pore-water velocity, and that all ions emerged from the soil cores well before 1 pore volume of drainage.

The general shape of the BTCs (steep leading edge followed by a more gradual trailing edge) was consistent with other workers (Pleysier and Juo 1981; Swarup et al. 1983; Phillips et al. 1988b). However, intermittent leaching resulted in a number of secondary peaks along the trailing edge (e.g. K^+). This behaviour is attributed to diffusion of ions from poorly-conducting or stagnant soil-pores into the primary conducting pores during the non-leaching periods. The amplitude of these secondary peaks diminished progressively with leaching because of decreasing amounts of the ions

Table 3. Amount of ions added to, and leached from, the soil core for each treatment.

Ion	Amount added (mg)				Amount in leachate (mg)				Amount in leachate (% of total applied)			
	K0	K50	K100	K300	K0	K50	K100	K300	K0	K50	K100	K300
K ⁺	0.00	307.72	615.44	1846.32	35.25	100.80	104.76	614.66		21%	11%	31%
NH ₄ ⁺	276.95	276.95	276.95	276.95	3.64	7.70	12.76	67.90	1%	3%	5%	25%
NO ₃ ⁻	0.00	0.00	0.00	0.00	72.16	63.48	82.15	51.53	*27%	*26%	*34%	*43%
Cl ⁻	0.00	276.95	553.90	1661.69	89.74	358.24	644.23	1636.25		97%	100%	93%
P	307.72	307.72	307.72	307.72	82.45	107.77	113.62	94.92	30%	35%	37%	31%

*Represents the combined recovery of NH₄⁺ and NO₃⁻.

within the stagnant pores with successive leaching events. This behaviour has been observed by other workers studying solute transport under intermittent leaching conditions (Addiscott et al. 1978; Rose et al. 1983; Ashraf et al. 1997; Al-Sibai et al. 1997).

The emergence of surface-applied solutes prior to 1 pore volume has generally been attributed to preferential flow within soil macropores (Nkedi-Kizza et al. 1982; Wildenschild et al. 1994; Bergstrom and Shirmohammadi 1999). Macropore flow is most obvious in solute transport studies using well-structured and/or well-aggregated soils at high pore-water velocities (Nkedi-Kizza et al. 1982; Williams et al. 1990; Wildenschild et al. 1994; Al-Sibai et al. 1997; Jensen et al. 2000). The similarity in fertiliser leaching behaviour in this study to that observed for macropore flow by other workers, suggests that a proportion of the applied fertiliser pulse primarily leached through the larger conducting soil pores and by-passed those pores not actively involved in transmitting water. This assumes that the total soil-water can be partitioned into mobile and immobile regions, which has been done in many water and solute studies involving undisturbed soil cores. Obviously some fertiliser had entered these poorly-conductive pores during the experiment (probably between leaching events), as indicated by the secondary peaks along the trailing edge of the BTCs. A qualitative estimate of the relative proportions of mobile and immobile soil-water is provided below.

The volume of 5000 ml represents an equivalent depth of about 8 cm, and should cause an increase in soil-water over the core length of about 0.10 cm³ cm⁻³. Interestingly, this value is very similar to the increase in soil-water contents

measured during each leaching event (i.e. $\Delta \theta_v \approx 0.10 \text{ cm}^3 \text{ cm}^{-3}$). It appears that much of the applied fertiliser leached from the soil cores was quickly transported throughout the soil core with the applied water, and moved with or immediately behind the wetting front. This analogy suggests that of the 0.2–0.25 cm³ cm⁻³ total soil-water content within the soil cores during leaching, only about 0.10–0.15 cm³ cm⁻³ appeared to be mobile, with the remaining 0.10 cm³ cm⁻³ being largely immobile. Transmission pores ($\approx 50 \mu\text{m}$ in diameter) empty at an applied suction of pF 1.6–2.0 (Rowell 1994), and much of the pore-water of the Podisol drained at similar pF values (Figure 3d). Interestingly, hydraulic conductivities predicted using the RETC code (van Genuchten et al. 1991), which analyses the hydraulic properties of unsaturated soils using a range of established soil-water retention and hydraulic conductivity functions, showed negligible water flow occurs at $\theta_v < 0.08\text{--}0.1 \text{ cm}^3 \text{ cm}^{-3}$ (Figure 3b). This amount of water may therefore be considered immobile, and it represents about 50% of the soil-water content during leaching events. This percentage attributed to immobile water is much higher than values of 5–15% reported for other coarse-textured soils (De Smedt and Wierenga 1984; Schulin et al. 1987; Bond and Wierenga 1990; Maraqa et al. 1997). These workers however conducted experiments using materials with an effective particle diameter of $< 1 \text{ mm}$, and it has been shown that the importance of macropore flow decreases with decreasing particle size (Al-Sibai et al. 1997; Maraqa et al. 1997). Furthermore, Phillips (2004) found no evidence of immobile water in horizontal constant absorption experiments using the $< 0.5 \text{ mm}$ size fraction of the Podisol soil.

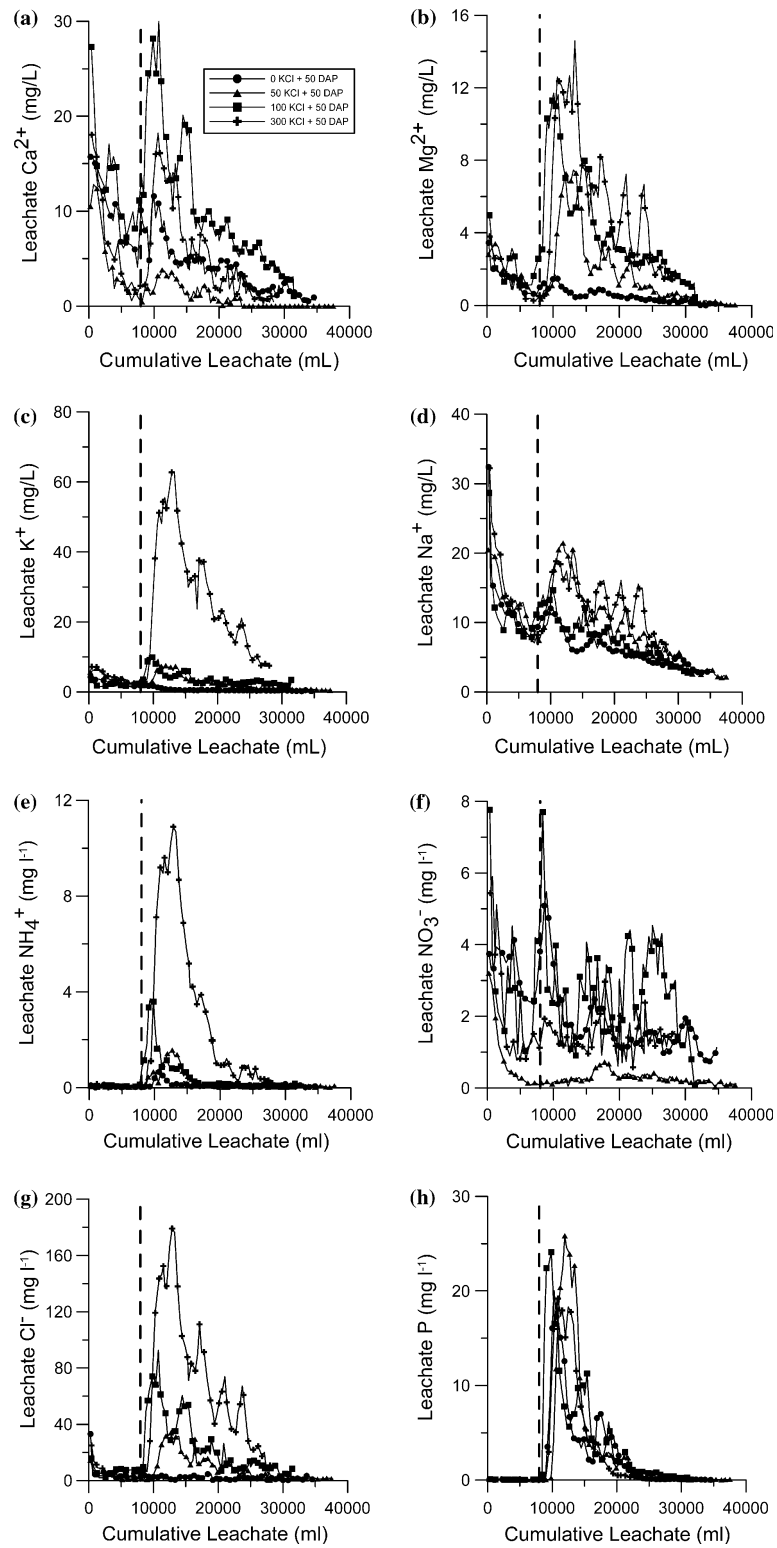


Figure 4. Distribution of (a) Ca^{2+} , (b) Mg^{2+} , (c) K^+ , (d) Na^+ , (e) NH_4^+ , (f) NO_3^- , (g) Cl^- and (h) P in the leachate from Core 1 (●), Core 2 (▲), Core 3 (■), and Core 4 (+). The dashed line represents when the solute pulse was applied to the soil cores.

As well as increasing the rate of water and fertiliser transport through soil, the presence of immobile water is likely to exacerbate fertiliser leaching by excluding a number of potential K^+ , NH_4^+ , and P sorption sites (Jensen et al. 2000). This effect would have been most pronounced during the irrigation and drainage periods when water movement through the soil core was occurring (Jensen et al. 2000). Whereas between each leaching event, opportunity existed for the soluble ions to diffuse into the immobile soil–water region, and to interact with cation exchange and other sorption sites (Nkedi-Kizza et al. 1982). For soils with a strong affinity for the applied ions, diffusion into immobile water regions should reduce the susceptibility of fertiliser ions to leaching (Nkedi-Kizza et al. 1982; Williams et al. 1990). However, the poor retention capacity of this Podsol soil has resulted in much of the ions remaining in the soil solution phase, with subsequent diffusion back into the main conducting pores (Figure 4).

The proportions of applied K^+ , NH_4^+ , Cl^- and P leached from each soil core are presented in Table 3. Essentially all of the applied Cl^- was leached from the soil cores, irrespective of KCl treatment. This was not unexpected given that Cl^- commonly behaves as a non-reactive chemical in soils with predominantly permanently negative charged colloids that do not exhibit anion exclusion, such as this Podsol. The high leachability of inorganic P (30% of applied) reported here is not typically found in P leaching studies due to the strong P sorption properties of most soils (e.g. Sanyal and De Datta 1991; Phillips 2002a, b). However, the extremely acid nature of this soil, coupled with poor sorption characteristics (Figure 2c), has resulted in even greater leachability of inorganic P than previously observed for other acid sandy soils (e.g. Gerritse 1989; Chen et al. 1996).

Approximately 21% (100 mg), 11% (105 mg) and 31% (615 mg) of the K^+ applied at the 50, 100 and 300 kg/ha rates leached from the soil cores, respectively (Table 3), and these percentage losses were similar to other K leaching studies (e.g. Pleysier and Juo 1981; Swarup et al. 1983; Phillips et al. 1988b). Furthermore, these results suggest that approximately 240, 550 and 1270 mg of applied K^+ was retained within a depth of 85 cm for the K50, K100 and K300 treatments, respectively. Although this soil retained much of the applied

K^+ , independent modelling (using the model by Phillips and Black 1991; data not presented) found that >50% of the applied K^+ was rapidly leached below a depth of 30 cm, which is typically below the root-zone of exotic pine trees. Furthermore, much of the sorbed K^+ was retained by non-specific exchange sites, and would be readily desorbed via cation exchange reactions, and potentially leached to groundwater.

Very little (1–25%) of the applied NH_4^+ was recovered in the leachate (Table 3). Cation exchange and nitrification are two mechanisms that could be responsible for the loss of this cation. Although not added to the soil, the presence of NO_3^- was detected in the leachate from all cores (Figure 4f). The BTCs were not characterised by a distinct maximum concentration as for the other ions (and found by other workers such as Pleysier and Juo 1981), and leachate concentrations prior to fertiliser application tended to be similar to those after application. Although nitrification as a mechanism for NH_4^+ loss cannot be ignored in this study, its contribution to inorganic N dynamics was not quantified in this study. It is possible that nitrification was occurring within the soil core to some extent, but this would be expected to be quite limited due to the rapid movement of applied NH_4^+ through the soil. Williams et al. (1990) found negligible nitrification of applied urea during macropore flow following surface application of cow urine. Furthermore, nitrification rates have typically been reported to be 1–2 mg kg⁻¹ per day (Phillips 2002a), and a period of time (2–3 days) to establish the nitrifying population may be required prior to nitrification commencing.

Cation exchange was considered to be the main mechanism responsible for the decrease in solution NH_4^+ concentrations. The Podsol was capable of sorbing limited amounts of this cation (Figure 1b). Also, under the K300 treatment, increased competition between NH_4^+ and K^+ caused NH_4^+ to remain in solution where it was rapidly leached (Figure 4e). Again, the absence of a characteristic NO_3^- BTC supports the conclusion that only negligible nitrification occurred as NH_4^+ leached through the soil. Work by Pleysier and Juo (1981) and Phillips (2002b) found that nitrification can be responsible for much of the loss of soluble NH_4^+ if the NH_4^+ can remain within the soil profile for extended periods of time.

Conclusions

Potassium and NH_4^+ was retained by this Podsol soil via cation exchange reactions. The magnitude of K^+ and NH_4^+ sorption was largely controlled by the soil's cation exchange capacity; however, the lower soil depths displayed a greater selectivity for these two cations than the surface depths. This was attributed to the higher organic matter contents of the surface depths because organic matter is known to exhibit a stronger preference for multivalent cations than monovalent cations. The greater preference of the soil for K^+ relative to NH_4^+ indicates that these two cations did not behave as true exchange analogs. Furthermore, this behaviour demonstrates the complex nature of cation reactions in soil, and highlights the need for continued research into one of the most fundamental chemical processes occurring in all soils.

Surface application of a combined KCl and DAP solution to the undisturbed Podsol soil cores (85 cm long) subjected to intermittent leaching, resulted in rapid loss of surface applied ions below a depth of 85 cm. Reactive and non-reactive ions appeared in the leachate simultaneously only after 0.25 pore volumes of drainage had been collected. The rapid leaching of ions was attributed to preferential flow, as only about 50% of the soil-water was regarded as being mobile. Secondary peaks were observed along the trailing edge of the ion BTCs due to ion diffusion into the soil matrix between each leaching event. Cation exchange was the dominant mechanism responsible for reducing applied K^+ and NH_4^+ concentrations in the leachate.

From an environmental perspective, the findings from this study suggest that significant fertiliser leaching to groundwater may occur in this exotic pine forest system, and may represent a serious risk for increased nutrient loadings to nearby sensitive marine environments. Further work in this project aims to quantify nutrient loadings in the groundwater and appropriate discharge points.

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